MIL-B-2331D
30 September 1985
SUPERSEDING
MIL-B-2331C
16 April 1969

MILITARY SPECIFICATION

BUCKLES, END BAR, WITH TONGUE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers end bar buckles with tongues.

1.2 Classification. Buckles shall be of the following types and classes as specified (see 6.2):

Type I - 1/2 inch without roller
Type II - 5/8 inch without roller
Type III - 5/8 inch with roller
Type IV - 3/4 inch without roller
Type V - 3/4 inch with roller
Type VI - 7/8 inch without roller

Class 1 - Brass
Class 2 - Steel

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-C-320	- Chromium Plating (Electrodeposited)
QQ-S-698	- Steel, Sheet and Strip, Low Carbon
QQ-W-321	- Wire, Copper Alloy
QQ-W-461	- Wire, Steel, Carbon, (Round, Bare, and Coated)
TT-E-529	- Enamel, Alkyd, Semigloss
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-676	- Boxes, Setup

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads

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DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

4-1-156 - Buckles, End Bar, with Tongue

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials. The materials shall conform to the referenced documents and requirements specified herein. Recycled material may be used (see 6.4).

3.2.1 Brass.

3.2.1.1 Wire. Brass wire shall conform to alloy No. 260 or 270, quarter-hard temper of QQ-W-321.

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3.2.1.2 Sheet or strip. Brass sheet or strip shall conform to copper alloy No. 260 or 268, quarter-hard temper of QQ-B-613.

3.2.2 Steel.

3.2.2.1 Wire. Steel wire shall conform to steel number 1006, 1008, or 1010, finish 1, annealed in process temper of QQ-W-461.

3.2.2.2 Sheet or strip. Steel sheet shall conform to cold rolled commercial quality, and steel strip shall conform to cold rolled skin-passed (No. 4) or dead-soft (No. 5) temper as required in QQ-S-698.

* 3.2.3 Enamel. Enamel shall comform to type II of TT-E-529. The color shall be black.

3.3 Construction. The construction of the buckles shall conform to the requirements specified herein and as shown on Drawing 4-1-156. All body joints shall be welded or brazed sound and smooth. The butting edges of the buckle rollers shall be butted evenly along the entire length and shall rotate freely without binding. All burrs, sharp edges, or slivers shall be removed prior to application of finish.

3.4 Finish.

3.4.1 Buckles, class 1. Except when the type VI, class 1 buckle is chromium plated (see 3.4.2), the class 1 buckles shall be vapor degreased and thoroughly cleaned. The buckles shall then be given one coat of clear commercial lacquer.

3.4.2 Buckles, class 1, type VI. When specified (see 6.2), the type VI, class 1 buckles shall be chromium plated in accordance with type I class 1, of QQ-C-320.

* 3.4.3 Buckles, class 2. The class 2 buckle shall be given a finish consisting of a phosphate treated zinc plate complying with type IV, classification Fe/Zn13 Sc3, of ASTM B 633. After phosphating and drying, surfaces shall be given a finish of enamel conforming to 3.2.3.

3.5 Marking for identification. The buckles shall be permanently and legibly marked with the manufacturer's identification of such known character as to be readily identifiable on the back surface of the buckle body. The identification marking shall be incorporated in such a manner that the front surface shall show no sign of penetration or unevenness nor shall it cause distortion on any part of the buckle.

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3.6 Workmanship. The finished buckles shall be free of rough or sharp edges, pits, pores, blowholes, cracks, sags, runs, wrinkles, pinholes, orange peel, dirt, dust, and areas of no film.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance. Any failure to conform to the quality assurance provisions of this document or any appearance failure shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or the applicable purchase documents.

4.4.2 Process examination. Process examination shall be conducted to see that accomplishment of specified requirements are in conformance to table I. Whenever nonconformance is noted, correction shall be made to the affected items and process.

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TABLE I. Process examination

Requirement, operation	Requirement paragraph
Cleaning and vapor degreasing of class 1 buckles	3.4.1
Phosphate treating and zinc plating of class 2 buckles	3.4.3

4.4.3 End item inspection. The lot shall be all the buckles of one type and class offered for inspection at one time. The sample unit for this inspection shall be one buckle.

4.4.3.1 Visual inspection. Examination shall be made of the buckles for defects in table II. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Category	
		Major	Minor
Finish	Finish missing or finish not as specified	101	
	Color not as specified		201
	Area of no finish or finish discolored		202

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TABLE II. Classification of defects - Continued

Examine	Defect	Category	
		Major	Minor
Finish (Continued)	Not uniform, not free from orange peel, sags, runs, pinholes, blushing, blisters, lint, dust, wrinkles, drops, streaks, or foreign matter		203
Construction and workmanship (all types and classes, as applicable)	Body joints not welded or brazed		204
	Butting edges of roller joint not even		205
	Roller binds		206
	Sharp edges, slivers, burrs, or flux residue not removed	102	
	Body, tongue, or roller malformed	103	
	Tongue mounted in reverse position	104	
	Tongue binds on end bar		207
	Tip of tongue protrudes beyond edge of roller	105	
Identification marking	Missing, incomplete, illegible, or incorrect		208
	Not on back surface of buckle		209
	Indication of penetration, unevenness, or distortion on face surface of buckle		210

4.4.3.2 Dimensional examination. Examination shall be made of buckles for compliance with dimensions specified. Any dimension that is not within the specified tolerance shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

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4.4.4 Packaging inspection. An inspection shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table III. The lot shall be all the containers offered for inspection. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE III. Packaging inspection

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application
Materials	Any component missing or damaged
Workmanship	Inadequate application of components such as twine or cord, adhesive, loose flaps, loose strapping, or tape banding
Content (interior and exterior)	Number is more or less than required

4.4.5 Palletization inspection. An inspection shall be made to determine that the palletization complies with section 5 requirements. Defects shall be scored as specified in the list in table IV. The lot shall be the number of palletized unit loads in the end item inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1, and the AQL shall be 6.5, expressed in terms of defects per hundred units.

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TABLE IV. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limit
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), buckles of one class and size only shall be unit packed in quantities specified in table V in a snug-fitting fiberboard box conforming to style RSC, type CF (variety SW) or type SF, class domestic, grade 200, of PPP-B-636. Each fiberboard box shall be closed in accordance with method II as specified in the appendix of PPP-B-636. When unit pack quantities other than those specified in table V are required, buckles of the same class and size only shall be unit packed in containers conforming to PPP-B-566, PPP-B-676, or PPP-B-636. The containers shall be sealed or closed in accordance with the applicable container document.

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TABLE V. Packaging and packing of buckles, end bar

Type and class of buckle		Quantity per fiberboard box	Quantity per shipping container
Type I	Class 1	1,800	10,800
	Class 2	2,000	12,000
Type II	Class 1	1,400	8,400
	Class 2	1,500	9,000
Type III	Class 1	1,100	6,600
	Class 2	1,200	7,200
Type IV	Class 1	1,400	8,400
	Class 2	1,500	9,000
Type V	Class 1	1,000	6,000
	Class 2	1,100	6,600
Type VI	Class 1	1,300	7,800
	Class 2	1,400	8,400

5.1.2 Commercial. The buckles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Unless otherwise specified (see 6.2), buckles of one class and size only, preserved as specified in 5.1, shall be packed in the quantities specified in table V in containers conforming to overseas type of PPP-B-601 or class 2, style 4 of PPP-B-621. Each container shall be closed and reinforced in accordance with the applicable container document and waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547.

5.2.2 Level B packing. Unless otherwise specified (see 6.2), buckles of one class and size only, preserved as specified in 5.1, shall be packed in the quantities specified in table V in a snug-fitting fiberboard shipping container conforming to style FTC, grade V3c or V3s of PPP-B-636. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

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5.2.3 Commercial packing. Buckles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), buckles packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The buckles are intended for use on miscellaneous leather and canvas equipment required by the Military.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.

b. Type and class of buckle required (see 1.2).

c. When a first article is required (see 3.1).

d. When type VI buckle requires chrome plating (see 3.4.2).

e. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).

f. Quantity per package and pack if other than specified (see 5.1.1, 5.2.1, and 5.2.2).

g. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instruction in all acquisition instruments regarding arrangements for inspection and approval of the first article.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
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Review activities:

Air Force - 82
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Preparing activity:

Army - GL

Project No. 5340-1626

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-2331D	Buckles, End Bar, With Tongue

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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